# Section 30: IPv6

Layer: L3 | Reference: Odom Vol1 Ch25-29: IPv6 Fundamentals, Addressing, Routers, Hosts, Routing | *Original study notes — CCNA 200-301*

IPv6 solves IPv4 address exhaustion with a vastly larger 128-bit address space and introduces new mechanisms for address assignment, neighbor discovery, and routing. The CCNA exam tests IPv6 addressing, SLAAC, and dual-stack/OSPFv3 concepts heavily.

## Key Concepts

- IPv6 addresses are 128 bits, written as eight groups of four hex digits separated by colons (e.g., 2001:0db8:0000:0000:0000:ff00:0042:8329)
- Leading zeros in a group can be omitted; one consecutive run of all-zero groups can be compressed to '::' (only once per address)
- Global Unicast Address (GUA): 2000::/3, globally routable, equivalent role to a public IPv4 address
- Link-Local Address (LLA): FE80::/10, automatically assigned to every IPv6 interface, used for on-link communication and routing protocol neighbor relationships, never routed off the local segment
- Unique Local Address (ULA): FC00::/7, the IPv6 analog to private IPv4 space (RFC 1918), not globally routable
- Multicast: FF00::/8 - IPv6 has no broadcast at all; multicast and anycast replace its functions entirely
- SLAAC (Stateless Address Autoconfiguration): host learns the prefix from a Router Advertisement (RA) and generates its own interface ID, commonly via EUI-64 (derived from the MAC address) or a randomized identifier for privacy
- Stateless DHCPv6: host still uses SLAAC for the address but queries a DHCPv6 server for additional info like DNS servers
- Stateful DHCPv6: DHCPv6 server assigns the full address, similar in spirit to DHCPv4
- DAD (Duplicate Address Detection): before using a new address, a host verifies no other host already has it
- NDP (Neighbor Discovery Protocol) replaces ARP: NS/NA (Neighbor Solicitation/Advertisement) resolve link-layer addresses; RS/RA (Router Solicitation/Advertisement) handle default gateway and prefix discovery
- Every IPv6 router interface needs a link-local address even if no global address is configured
- OSPFv3 is the IPv6-routing version of OSPF; same area/LSA concepts apply, configured per-interface with 'ipv6 ospf' rather than network statements
- Dual-stack means running IPv4 and IPv6 simultaneously on the same interface/device during migration

## Command Reference

| Command | Purpose |
|---|---|
| `ipv6 unicast-routing` | Globally enable IPv6 routing on a router (off by default) |
| `ipv6 address <addr>/<prefix>` | Manually assign an IPv6 address to an interface |
| `ipv6 address autoconfig` | Enable SLAAC on an interface to auto-generate an address |

| Command | Purpose |
|---|---|
| `ipv6 enable` | Enable IPv6 processing on an interface (auto-generates a link-local address) without a global address |
| `show ipv6 interface brief` | Quick view of IPv6 addresses and interface status |
| `show ipv6 route` | Display the IPv6 routing table |
| `show ipv6 neighbors` | Display the IPv6 neighbor table (NDP cache, analog to ARP table) |
| `ping ipv6 <address>` | Test IPv6 reachability (or just 'ping <ipv6-addr>' on most platforms) |
| `ipv6 ospf <process-id> area <id>` | Enable OSPFv3 directly on an interface (no network statement used) |

## Configuration Walkthrough

### Manually configure a global IPv6 address and enable routing

```
R1(config)#ipv6 unicast-routing
R1(config)#interface GigabitEthernet0/0
R1(config-if)#ipv6 address 2001:db8:1:1::1/64
R1(config-if)#no shutdown
```

### Enable SLAAC on a LAN-facing interface

```
R1(config)#interface GigabitEthernet0/1
R1(config-if)#ipv6 address autoconfig
R1(config-if)#no shutdown
```

### Enable OSPFv3 directly on interfaces

```
R1(config)#interface GigabitEthernet0/0
R1(config-if)#ipv6 ospf 1 area 0
R1(config)#interface GigabitEthernet0/1
R1(config-if)#ipv6 ospf 1 area 0
```

## Worked Scenario

**Q: A host using SLAAC has a valid-looking IPv6 address but can't reach anything beyond the local subnet. What's the most common cause?**

Usually a missing or malformed Router Advertisement. SLAAC needs the RA to supply the prefix and, critically, the 'default router' information; if the RA doesn't carry that (or RA suppression is misconfigured on the router), the host can self-assign an address but has no default gateway for off-link traffic. Verify with 'show ipv6 interface' on the router to confirm RAs are being sent, and check the host's discovered default gateway.

## Common Mistakes / Gotchas

- Forgetting 'ipv6 unicast-routing' globally - without it, a router won't forward IPv6 packets between interfaces even if addresses are configured
- Assuming IPv6 has a broadcast address like IPv4 - it does not; multicast and anycast take over those roles entirely
- Confusing link-local (FE80::/10, never routed) with unique local (FC00::/7, private-equivalent but technically routable within an org)
- Forgetting every interface needs SOME IPv6-enabled state (even just 'ipv6 enable') to generate a link-local address before NDP/OSPFv3 will work

## Exam Tips

★ *Practice compressing/expanding addresses quickly: rule is one '::' run per address, leading zeros droppable per group*

★ *Know the address type ranges cold: 2000::/3 GUA, FE80::/10 link-local, FC00::/7 ULA, FF00::/8 multicast*

★ *NDP replaces ARP: NS/NA = address resolution, RS/RA = gateway/prefix discovery*